PINGU TECH
Where Ideas March Forward

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Analysing Report

For Integrated Medical Booking System

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1. Cover Letter

PINGU TECH
Where Ideas March Forward

84 Punggol Way, #01-86/87
Singapore 829911

2nd February 2026
Mr. Direk Tor Delivery Director

SUBMISSION OF ANALYSING REPORT AR-1223:
MEDICAL INC. INTEGRATED MEDICAL BOOKING SYSTEM

Dear Mr. Direk Tor,

Attached is Analysing Report AR-1223 for the Medical Inc. Integrated Medical Booking System project. This report provides a technical and risk assessment of the Project Specification (Version 1.0) to ensure we deliver successfully and profitably.

Key Outcomes & Objectives

- **Efficiency:** Automate scheduling across 3 locations to achieve a 40% reduction in administrative workload.
- **Integrity:** Ensure 100% data consistency and Zero system-caused double-bookings.
- **Access:** Provide patients with 24/7 self-service booking and automated notifications.
- **Compliance:** Deployment meeting all PDPA/MOH requirements.

We recommend immediate approval to proceed with proposal finalization.
Please note the client's submission deadline is **10 March 2026.**

Regards,

SP

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Project Lead

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2. Executive Summary

2.1. Context

Medical Inc., a Singapore-based family medicine operator with three clinic locations, currently manages appointment scheduling through entirely manual, paper-based processes. This approach has resulted in:

- Recurring double-booking incidents across three clinic locations
- Inconsistent patient records due to the absence of inter-clinic data synchronization
- High administrative overhead, limiting staff capacity for direct patient care
- Inability to scale operations to meet increasing patient demand

This report analyses the Project Specification to determine whether we can deliver successfully, profitably, and with acceptable risk.

2.2. Project Objectives

The Medical Inc. Project Specification defines three high-level SMART objectives that our proposed solution must satisfy:

2.2.1. Objective 1: Multi-Clinic System Deployment

- **Specific:** Develop and deploy a fully functional, web-based appointment booking platform that supports at least 3 clinic locations. The platform's core features will include real-time display of doctor availability, patient self-service for booking/cancelling/rescheduling appointments 24/7, automated email/SMS confirmations and reminders, and a centralized, user-friendly interface
- **Measurable:** Success is measured by the full implementation of all specified features: real-time multi-location availability display, 24/7 booking/cancellation/rescheduling, automated email/SMS notifications, and a user-friendly interface, as validated by stakeholder sign-off
- **Attainable:** The scope is well-defined, focusing on proven scheduling technology, and is achievable with a dedicated development team and active clinic cooperation for integration and testing
- **Relevant:** This directly addresses the core business need to modernize appointment booking, improve patient access and experience, and serve multiple locations from a single, efficient system
- **Time-bound:** Full deployment must be completed within 6 months of the project start date.

2.2.2. Objective 2: Operational Efficiency & Administrative Burden Reduction

- **Specific:** Reduce the administrative workload by automating the appointment scheduling process across the 3 clinic locations.
- **Measurable:** Achieve a 40% reduction in the total time spent by staff on manual scheduling tasks (e.g. phone calls, calendar management, sending reminders), measured by comparing pre-launch and post-launch time-tracking logs.
- **Attainable:** A 40% reduction is realistic as the platform automates the most time-consuming tasks. Baseline data will be collected from current manual processes to ensure the target is data-driven.
  Relevant: Reducing manual tasks frees up administrative staff for higher-value work, lowers operational costs, and minimizes errors, directly supporting MEDICAL INC's initiative for operational efficiency enhancement.
- **Time-bound:** This reduction must be demonstrated within 4 months of the platform's official launch.

2.2.3. Objective 3: Cross-Clinic Data Integrity & Synchronization

- **Specific:** Ensure absolute accuracy and real-time synchronization of all appointment data across the booking platform and the integrated systems at all 3 clinic locations.
- **Measurable:** Maintain 100% consistency of appointment records (time, date, doctor, patient) in real-time. Achieve zero (0) reported incidents of double bookings or data mismatches caused by a system error after launch.
- **Attainable:** This is achievable through robust database architecture (e.g. using transactional operations and real-time sync) and thorough pre-launch testing of concurrency and integration points.
- **Relevant:** Data integrity is fundamental to clinical operations, patient trust, and resource utilization. Preventing double-booking errors is critical for the system's credibility and supports the key business initiative of inter-clinic data consistency.
- **Time-bound:** This standard of integrity and synchronization must be maintained from the moment of platform launch and continuously thereafter.

The solution shall be hosted on a Singapore-resident cloud infrastructure (AWS/Azure/Google Cloud) with a multi-tier architecture to ensure high availability and scalability.

2.3. Solution Value

If we deliver this project successfully, the following outcomes will be achieved for the client:

Metric	Current State	Post-Project State

Double-booking incidents	4–6 per week	Zero (system-caused)
Scheduling admin time	3.5 hours/clinic/day	2.1 hours/clinic/day
Patient booking access	Clinic hours only	24/7 self-service
Data consistency	Siloed per clinic	100% synchronised
PDPA compliance	Manual, inconsistent	Automated encryption, audit, retention

For our organisation, successful delivery provides:

- A referenceable healthcare client in the Singapore primary care market, a sector with high replication potential and low vendor saturation.
- Predictable revenue through milestone-linked payments that protect cash flow.
- Reusable technical assets: The booking engine, reservation lock logic, and multi-clinic synchronisation pattern are transferable to future engagements.
- Reputational positioning: Delivering a 40% efficiency improvement and zero-defect data integrity positions us as an outcome-driven partner, not merely a software contractor.

2.4. Proposed Actions

The following actions are required to position our bid competitively while protecting our commercial and delivery interests:

Step	Action	Owner
1	Review and approve this Analysing Report	Internal Leadership

2	Submit formal clarifications to Medical Inc.: • Request anonymised production data sample • Propose contractual definition of "real-time" (≤ 3 seconds)	Business Analyst / Technical Lead
3	Finalise commercial response: • Scope SingPass as separate option with GovTech dependency clause • Ensure contingency buffer is protected for genuine unforeseen risks	Commercial Lead / Project Lead
4	Design technical mitigations for inclusion in proposal: • Pilot data migration script (Sprint 1) • API fallback mechanisms (e.g. "Pay at Clinic" if gateway down) • Low-complexity UX workflow for clinic staff	Solution Architect / UX Lead
5	Submit proposal to Medical Inc.	Project Lead

2.5. Why It Matters

This project matters because manual processes in healthcare are no longer acceptable. Medical Inc. has recognised that its legacy approach is actively impeding its mission to provide accessible, efficient, and compassionate family healthcare.

For us, this project represents a low-risk, high-value opportunity to:

- Enter or strengthen our position in the Singapore healthcare technology market.
- Deliver measurable, contractually verified outcomes.
- Build reusable assets that reduce cost-to-deliver on future engagements.

We recommend immediate approval to proceed with proposal finalisation and contract negotiation.

3. BCIC (Breakdown / Clarify / Interpret / Categorize)

IRC Item	Reqrs	Breakdown	Clarify	Interpret	Categorize
Imperative Language Weakness	BO-REQ-01 , 03, 04, 08	Essential reqrs used weak verbs (“should”), reducing enforceability.	BA met Project Lead; explained contractual difference between “should” vs “shall”. Agreement: Essential = mandatory.	All Essential reqrs rewritten using “shall” per IEEE standard.	Updated across Functional & Non-Functional sections.
High System Availability Unrealistic	BO-REQ-09	“Available at any time” operationally impossible; not measurable.	BA presented SLA model to Ops Lead. Agreed 99% monthly uptime acceptable.	Defined measurable uptime threshold and maintenance notice rules.	NFR-AV-01: ≥99% monthly uptime. NFR-AV-02: Maintenance ≥48hr notice, outside 7am–10pm.
Automated Notifications Ambiguous	BO-REQ-06	Trigger timing and workflow undefined.	BA used event-trigger flow diagram with admin stakeholders.	Defined immediate confirmation, 24h reminder, and instant cancellation notice.	FR-NTF-01: Confirmation immediate. FR-NTF-02: Reminder 24h prior. FR-NTF-03: Cancellation notice immediate.
Doctor Availability Management Ambiguous	BO-REQ-10	Approval rules and emergency handling unclear.	BA conducted scenario walkthrough (planned vs emergency leave).	Defined ≥48h planned notice, emergency approval, mandatory patient notification.	FR-DOC-01: ≥48h planned notice. FR-DOC-02: Emergency requires admin approval. FR-DOC-03: Affected patients auto-notified.

Waiting List Incomplete	BO-REQ-05	FCFS stated but no response window or expiry rule.	BA modelled queue lifecycle with admin staff.	Defined timestamp ordering and 2-hour confirmation expiry	FR-WL-01: Timestamp-based ordering. FR-WL-02: 2-hour response window before next patient notified.
Payment Logging vs No Storage Conflict	TR-REQ-07	Conflict between “no payment data storage” and “log receipts”.	BA clarified with Finance & Compliance using payment data flow model.	Prohibited storage of sensitive card data; allowed transaction metadata logging.	NFR-SEC-01: No PAN/CVV storage. FR-PAY-01: Store transaction ID, amount, timestamp only.
SingPass Optional Dependency	BO-REQ-01 , TR-REQ-08	Optional integration but core dependency unclear.	BA confirmed with Regulatory that system must function without SingPass.	Defined optional authentication; core booking independent of SingPass.	FR-AUTH-01: Optional SingPass support. FR-AUTH-02: Core booking independent of SingPass availability.
Role Capability Ambiguity	BO-REQ-02 , 10	Possible privilege overlap between nurse/admin.	BA facilitated role-permission matrix workshop.	Defined explicit Role Access Matrix (CRUD per role).	Functional → Access Control (Appendix Model)
Referral Processing Feasibility	BO-REQ-11	Workflow incomplete; approval checkpoint missing.	BA clarified with doctors.	Defined nurse drafting + mandatory doctor authorization before release.	FR-REF-01: Nurses draft. FR-REF-02: Doctor digital authorization required before release.
Admin & Monitoring Scope Broad	TR-REQ-11, 12	Monitoring requirement overly broad.	BA aligned with Technical Lead on minimal viable monitoring scope.	Defined external uptime monitoring and mandatory admin audit logging.	NFR-MON-01: External uptime monitoring. NFR-MON-02: Audit logs for admin actions.

Real-Time Terminology Ambiguous	BO-REQ-03	“Real-time” undefined and not testable.	BA defined measurable latency with stakeholders.	Defined maximum propagation delay under normal conditions.	NFR-RT-01: Updates propagate ≤3 seconds under normal conditions.

4. Incongruous Requirements Checklist (IRC)

IRC Type	Observation	Incongruity Identified	Correction / Rationale	Requirement
Imperative	Essential requirements need strong obligation language	Several Essential requirements currently use weak verbs like “should” instead of strong obligation terms	All Essential requirements shall use “shall” to indicate non-negotiable obligations, ensuring enforceability and regulatory compliance	BO-REQ-01, BO-REQ-03, BO-REQ-04, BO-REQ-08
Continuance	N/A	No excessive use of linking words or phrases used	Requirements do not use words like “Below”, “Following”, or “In particular” no correction needed	All requirements
Contextual	High system availability wording too broad	Unrealistic	“Available at any time” is operationally unrealistic. Uptime ≥ 99% monthly scheduled maintenance windows defined outside peak hours (7am–10pm)	BO-REQ-09
Contextual	Automated appointment notifications	Ambiguous	Timing and channel assumptions need clarification. Define triggers for confirmations, reminders,	BO-REQ-06

			and cancellations to maintain workflow continuity	
Contextual	Doctor availability management	Ambiguous	Schedule changes impact patients and workflows Define lead time, approval rules, and notification mechanism to maintain continuity	BO-REQ-10
Contextual	Waiting list management	Suspected incompleteness	Tolerable requirement, but process must ensure queue order preserved, notifications sent promptly	BO-REQ-05
Contextual	Payment gateway integration	Unverifiable with customer policy	External dependency Logs stored, sensitive data not persisted, API transactions handled securely	TR-REQ-07
Contextual	Optional SingPass login	Unverifiable with customer policy	External dependency SingPass login may be used optionally without blocking core functionality	BO-REQ-01, TR-REQ-08
Contextual	Role capabilities vary	Ambiguous	Potential misinterpretation if role permissions unclear Contextual review ensures correct dashboard mapping to user types	BO-REQ-02, BO-REQ-10
Contextual	Referral processing	Substantively invalid	Desirable requirement. Conditional on doctor approval. Ensure process is feasible and compliant	BO-REQ-11

Contextual	Administration & monitoring tools	Substantively invalid	Desirable.Ensure maintainability without affecting essential operations	TR-REQ-11, TR-REQ-12
Contextual	Real-time terminology	Ambiguous	Ambiguous term in testing Define acceptable latency thresholds and system-wide consistency to clarify	BO-REQ-03

5. Project Repository (Standardized Workspace)

Access Link: https://github.com/Pingu-Tech/Medical-Inc-Integrated-Booking-System.git
To support the delivery of the Integrated Medical Booking System within our established 6-month timeline, our team has provisioned a standardized, high-integrity project workspace. This repository is the central authority for all analyzed requirements and architectural blueprints, designed to minimize friction as we move towards the finalization of the Software Requirements Specification (SRS).

Site Organization
The repository follows a tiered structure that separates client-facing compliance documentation from our internal technical scaffolding. This ensures that all members of the project team can access relevant information without compromising system security or project focus. More information about the repository structure and its justifications can be found in the repository's *README.md.*

6. Account Management & Strategic Alignment

As acting Account Manager, my focus is on maximizing the lifetime value of the Medical Inc. partnership while de-risking our current 6-month delivery window.

6.1. Customer Relationship & Scope

I have established direct channels between clinical staff and our development team to validate uncertainties early. I act as a strategic buffer, ensuring stakeholders feel heard while shielding our engineers from "noise".

6.2. Contractual Success Metrics

To prevent scope creep, I have anchored project acceptance to measurable KPIs (100% Data consistency, 40% admin reduction). These are defined as “contractual gates” for milestone sign-off.

6.3. Shared commitment

I have secured a dedicated training budget and a milestone-based payment schedule (30% front-loaded / 20% at-risk). This ensures the client is operationally and financially invested in a successful rollout.

7. Risk Analysis

This analysis identifies potential threats or hindrances towards the successful delivery of the Integrated Medical Booking System.

7.1. Third-Party Integration

- 7.1.1. If we utilize external APIs for critical functions, like SingPass for authentication and Payment Gateways for transaction processing, external service outages or unexpected maintenance could render the booking system unusable, hindering clinic operations.
- 7.1.2. Impact: High
  - 7.1.2.1. If SingPass were to fail, patients cannot authenticate to book/reschedule appointments.
  - 7.1.2.2. If Payment Gateways fail, patients will be unable to make payments for their visit, causing inconvenience to patients.
- 7.1.3. Mitigation Strategy
  - 7.1.3.1. Enabling manual verifications/payment options during outages.

7.2. Migration Complexity

- 7.2.1. The project aims to replace manual appointment processes with a centralized database, with the assumption of “minimal transformation” of data. However, manual records may contain “dirty” data like missing/incorrect information due to human error, and importing the data will cause mass transaction failures.
- 7.2.2. Impact: High
  - 7.2.2.1. Continuous failed migrations or turning to manual labor of fixing the discrepancy might threaten the 6-month delivery timeline.
- 7.2.3. Mitigation Strategy
  - 7.2.3.1. Execute mock migrations on subsets of data monthly to bimonthly to identify problems early for rectifying.

7.3. System Adoption

7.3.1. The new system will be deployed across the 3 existing physical clinic locations, and each staff possesses varying levels of technical proficiency. If the new system is too technical and complex, non-technical staff may revert to manual workarounds (e.g back to writing on paper) or take substantial time tending to each patient.

7.3.2. Impact: Medium-High

7.3.2.1. Data inconsistencies can happen between physical clinics and the centralized database.

7.3.2.2. The target of 40% reduction in administrative tasks might be unable to be achieved.

7.3.3. Mitigation Strategy

7.3.3.1. Involve a representative expected to often utilize the system to co-design to ensure the digital workflow and complexity matches physical reality.

7.4. Regulatory Compliance (PDPA/MOH)

7.4.1. The healthcare system handles sensitive patient data and must strictly adhere to PDPA and MOH guidelines. With regulations ever-changing, any changes in guidelines during the 6 month development cycle could induce additional core logic changes.

7.4.2. Impact: Medium

7.4.2.1. Potential delays to refactor/redesign codes to comply with new guidelines

7.4.3. Mitigation Strategy

7.4.3.1. Decouple compliance logic (e.g. Data Retention Policies, Consent Logging) into a dedicated Compliance Service. This ensures that regulatory updates only require changing one module, not the entire application.

8. Sign-Off

This Analysing Report has been prepared to inform Pingu Tech leadership and the bid team of our technical, commercial, and risk assessment of the Medical Inc. Integrated Medical Booking System Project Specification. It reflects a complete BCIC and IRC review of all 23 business and technical requirements, identifies critical risks, and proposes specific, actionable mitigations.

Sign-Off Statement

We, the undersigned project team members, confirm that:

1. Analysis is complete. All requirements in the Medical Inc. Project Specification (Version 1.0, 02 February 2026) has been reviewed and assessed against the BCIC and IRC frameworks.
2. Findings are accurate. The incongruities, ambiguities, and risks documented in this report faithfully represent our collective analysis and have not been suppressed or understated.
3. This document is ready for leadership review. We recommend immediate escalation to the Delivery Director and Head of Projects for approval to proceed with proposal finalisation and contract negotiation.

Signatures

Role	Name	Signature	Date
Project Lead	Si Pei	SP	30/1/26
Research & Development / Account Manager	Galvin	Galvin	28/1/26
Cybersecurity & Privacy Expert	Shaqeel	Shaq	28/1/26
Business Analyst	Randy	Rand	29/1/26
Backend Developer	Alyssa	A	30/1/26
Quality Assurance Officer	Reuben	Reuben	29/1/26

Next Step Approved By:

Role	Name	Signature	Date
Delivery Director	Direk Tor		
Head of Projects	John Projek		

Addenda A

AI Declaration

We declare that generative AI tools (specifically Gemini 3 Pro) were used in the preparation of this project specification.

- **Purpose of Use**

  The AI tool was used for ideation, refining sentence structure and formatting.

- **Verification**

  All AI-generated content has been reviewed, fact-checked, and modified by the team members. We take full responsibility for the accuracy and the originality of the content submitted.

- **Data Privacy**

  No confidential or personally identifiable information regarding the client or team members was input into the AI tool.

Addenda B

Plagiarism Declaration

We hereby declare that this Project Specification and all accompanying documentation are the original work of the team members listed below.

We confirm that:

No Plagiarism: No part of this document has been copied from other sources (including other teams, online repositories, or unauthorized materials) without proper attribution.

Integrity: This work has been prepared solely by the team members for the purpose of the Medical Inc. Unified Clinic Management System project.